# Psychological and Functional Impacts of Climate Change Exposure on Adolescents: A Systemic Review


## Abstract

Background:
Climate change is an escalating global threat with significant implications for adolescent mental health. Both direct exposure to climate-related disasters and indirect exposure through awareness of environmental degradation may contribute to psychological distress. Adolescents represent a particularly vulnerable population due to ongoing emotional, cognitive, and social development. This systematic review examined the psychological manifestations and functional impacts of climate change exposure among adolescents and explored factors that may moderate these outcomes.

Methods:
A systematic review was conducted in accordance with PRISMA guidelines. PubMed, Cochrane Library, and ClinicalTrials.gov were searched for studies published between January 2000 and April 2026. Eligible studies included adolescents aged 10–19 years exposed to direct climate-related hazards or indirect climate-change awareness. Primary outcomes included anxiety, depression, climate anxiety, post-traumatic stress symptoms, suicidality, sleep disturbances, school functioning, and coping mechanisms. Study quality was assessed using the Newcastle-Ottawa Scale, CASP checklist, and Cochrane Risk of Bias tool, as appropriate.

Results:
Fifteen studies met inclusion criteria from 59 identified records. Evidence consistently demonstrated associations between climate-related experiences and adverse psychological outcomes, including anxiety, depressive symptoms,

hopelessness, post-traumatic stress symptoms, emotional distress, and suicidality. Direct exposure to climate-related disasters such as floods, mudslides, and wildfires was associated with increased rates of depression, PTSD, and suicidal ideation. Indirect exposure through climate awareness and future environmental concerns was linked to climate anxiety, sleep disturbances, and depressive symptoms independent of generalized anxiety. Several longitudinal studies identified bidirectional relationships between climate awareness and psychological distress. Moderate climate concern was associated with increased environmental engagement and adaptive coping, whereas severe eco-anxiety was linked to functional impairment and poorer mental health outcomes.

Conclusion:
Eco-anxiety is an increasingly important mental health concern among adolescents. Both direct climate-related experiences and anticipatory concerns regarding future environmental threats contribute to psychological distress. While moderate climate concern may promote adaptive engagement, excessive eco-anxiety may impair functioning and increase psychiatric vulnerability. Early identification, resilience-building interventions, and developmentally informed mental health strategies are needed to support adolescents facing the psychological consequences of climate change.

# Introduction

The escalating climate crisis is an unprecedented threat to the mental health and well-being of the current and future generations[8]. Adolescents are vulnerable to both direct exposure through acute climate hazards and indirect exposure via climate news and the social representation of existential environmental threats[8]. Globally, a foundational survey of 10,000 young people found that nearly 60% report being "very" or "extremely" worried about climate change, with many indicating that these feelings negatively impact their daily functioning[6,14]. This distress is felt globally, compounding into a ‘polycrisis’ in climate-vulnerable nations like Colombia and Bangladesh, where environmental degradation intersects with poverty, food insecurity, and the loss of ancestral lands[6].

Adolescence represents a critical developmental window defined by rapid physical, social, and cognitive changes that heighten sensitivity to global issues[3,13]. During this period, brain maturation and hormonal fluctuations amplify emotional reactivity, making this demographic particularly sensitive to the existential threat of a warming world[14]. As youth develop the capacity for abstract thought and hypothetical future-planning, they become increasingly aware of the long-term consequences of climate change[11]. Unlike many adults, adolescents often have a lower tolerance for dissonance between environmental knowledge and action, which can drive intense frustration when they perceive a lack of societal response[11]. Younger adolescents are in a critical period where their developing emotion regulation skills dictate whether they view climate change as a manageable challenge or a paralyzing threat[8].

The psychological response to climate change manifests through a complex spectrum of emotions, including anger, hopelessness, guilt, and sadness[14]. Climate-related worry is a unique predictor of depression and sleep disturbances, independent of generalized anxiety levels[8]. In severe cases, this distress triggers impairments in concentration, school performance, and social relationships[3]. In high-risk settings, the cognitive overload resulting from repeated disasters can lead to temporal discounting, where the immediate psychological burden of current floods or heatwaves impairs the capacity for long-term adaptation planning[6]. Marginalized and clinical populations face heightened risks; environmental degradation has been linked to spiritual disharmony and increased suicide risk among Indigenous youth and hospitalized psychiatric patients[5].

When youth feel that governments and older generations are failing to act on climate change with urgency, a psychological phenomenon known as "institutional betrayal" manifests[1]. Their emotional distress is severely compounded, often paralyzing them with anxiety. Moving from this state of paralysis to mobilizing action depends heavily on the adolescent's internal coping mechanisms and their perception of adult authority[14]. While individual pro-environmental habits are positive, focusing solely on personal, isolated actions can heighten anxiety by making the individual feel entirely responsible for a massive, systemic crisis[11]. To buffer against this despair, youth need meaning-focused strategies like positive reappraisal alongside collective action frameworks that shift the burden to a shared group effort[11]. Ultimately, when adults and institutions transparently discuss mitigation efforts, they restore a sense of institutional legitimacy, validate youth emotions, and model the very prosocial behaviors required to build collective trust and maintain durable optimism[11].

Despite a growing body of evidence, there is a critical need to synthesize how diverse exposures and internal/external moderators interact across geographically fragmented global contexts to shape the adolescent experience[4].

This systematic review explores the psychological manifestations and functional impacts of direct and indirect climate-change exposure on adolescents globally, and the mechanisms and perceived institutional responses that moderate these outcomes

The objective of this review is to evaluate the psychological and functional consequences of climate change on the adolescent demographic by synthesizing international evidence from diverse backgrounds.

This review seeks to identify the pathways of distress and the systemic factors that promote resilience during this pivotal stage of life.

# Review

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## Methods

This systematic review follows the PRISMA guidelines to evaluate the psychological manifestations and functional impacts of direct and indirect climate-change exposure on adolescents globally.

## Literature search

A comprehensive search for relevant literature was conducted to identify studies investigating the mental health, functional, and behavioral responses of adolescents to climate-related stressors.

Two electronic databases and one register - PubMed, Cochrane Library, and ClinicalTrials.gov - were systematically searched for peer-reviewed studies published between January 1, 2000, and 2026.
A detailed overview of the search strategy is presented in table 1.

## Search Strategy

**Studies found through Data Bases: 40**
**Studies found through Citation Search: 19**

| SEARCH STRATEGY | DATABASE/ REGISTERS | NUMBER OF STUDIES BEFORE AND AFTER FILTERS | FILTERS APPLIED |
|---|---|---|---|
| (("climate change" OR "global warming" OR "extreme weather" OR flood* OR wildfire* OR drought* OR "natural disaster*") AND (("climate anxiety" OR "eco-anxiety" OR "climate distress" OR "psychological manifestation*" OR "functional impact*" OR "mental health" OR "psychological distress") | PUBMED | 33/3660 | Full text, Clinical Study, Clinical Trial, Comparative Study, Controlled Clinical Trial, Equivalence Trial, Evaluation Study, Interview, Observational Study, Randomized Controlled Trial, Validation Study, Video-Audio Media, Webcast, English, Humans, Adolescent: 13-18 years, from 2000 - 2026 |
| "Climate Anxiety" OR "Eco-anxiety" OR "Climate Distress" AND adolescent* OR youth OR teen | clinicaltrials.gov | 2/5 | Completed |
| (adolescent* OR youth OR "young people" OR teen* OR pediatric*) AND ("climate change" OR "global warming" OR "extreme weather" OR flood* OR wildfire* OR drought*) AND ("climate anxiety" OR "eco-anxiety" OR "climate distress") | Cochrane Library | 5/6 | From 2000-2026, English |
| | | | |

## Eligibility Criteria

Studies were eligible for inclusion if they focused on humans aged 10–19 who experienced either direct climate hazards or indirect exposure via news and awareness. Validated scales were used to measure outcomes such as psychological well-being, functional impacts, temporal discounting and mental health outcomes. This specific phase of the systematic review, involving the finalization and application of these criteria, was conducted between March 25th and April 16, 2026.

Studies were excluded if they involved populations outside the target adolescent age range (10–19 years), or examined stressors unrelated to climate change. We also omitted research strictly limited to physical health outcomes, pharmacological interventions, or workplace-only exposures. Furthermore, secondary sources such as literature reviews, editorials, and commentaries were excluded to prioritize primary clinical data published in English since 2000.

A clear overview of Inclusion and Exclusion Criteria is given in table 2.

# Systematic Review: Inclusion and Exclusion Criteria

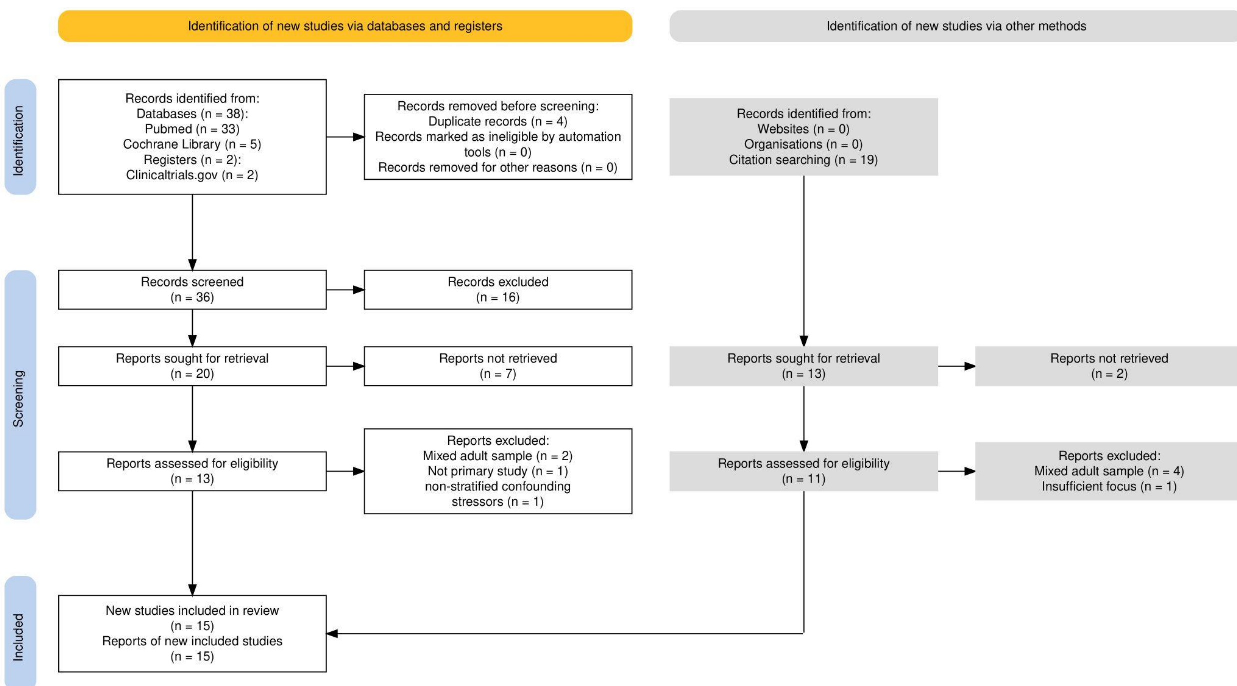
**Methodological Note:** Priority is given to studies using validated scales (e.g., PHQ-9, GAD-7, and Climate Change Anxiety Scale) over single-item self-reports.

| CATEGORY | INCLUSION CRITERIA | EXCLUSION CRITERIA |
|---|---|---|
| **Population** | • Humans only<br>• Adolescents (aged 10-19)<br>• Global representation (climate-vulnerable settings)<br>• Indigenous populations in high-risk zones | • Non-human/animal models<br>• Adults over age 19<br>• Pre-existing unrelated clinical conditions<br>• Stressors unrelated to climate |
| **Exposure** | • **Direct Exposure:** Personal experience with climate hazards (floods, heat, wildfires, drought, etc.)<br>• **Indirect Exposure:** Climate news/media awareness, "fear for the future," and anticipated losses | • General pollution without climate link<br>• Occupational/workplace-only exposures |
| **Outcome** | • **Psychological Well-being:** Clinical distress (GAD-7 for anxiety), depression, and eco-anxiety symptoms<br>• **Temporal Discounting:** Impact on long-term planning and short-term reward preference<br>• **Functional Impacts:** Impairment in daily life (school, sleep, eating, social relationships)<br>• **Behavioral Engagement:** Pro-environmental behaviors (PEB), political participation, or career shifts | • Purely physical health outcomes<br>• Medical or pharmacological interventions |
| **Study Design** | • **Clinical Studies:** Cross-sectional studies, cohort, observational, and prospective studies | • Papers, Editorials, commentaries, or literature reviews that do not present new primary data |
| **Publication** | • Published in the English language<br>• Published since 2000 (last 10-20 years)<br>• Full text available | • Non-English publications<br>• Published before 2000<br>• Abstracts or posters only |

# Results

## Prisma Diagram

An initial pool of 59 records was identified through database searches (n=40) and citation searching (n=19) and uploaded to Rayyan for screening. Ultimately, 15 studies were included in the review. During the eligibility phase, several reports were excluded for failing to meet the strict inclusion criteria, most notably due to mixed adult samples (n=6) that did not isolate the target adolescent population. Other exclusions were attributed to a lack of primary data, insufficient thematic focus, or non-stratified confounding stressors..



## Data extraction and management

To gather the relevant data, a standardised data extraction form was created to ensure the same facts were collected from every study, such as the authors, study goals, and results. Two authors worked together to extract the data, and then a third senior author checked the work to ensure accuracy. Any disagreements on specific details were resolved through discussion with the senior author until a consensus was reached.

## Study characteristics

The following table, Table 3, provides a comprehensive overview of the 15 studies included in this review, detailing their geographical contexts, methodology, and primary outcomes. The included research spans diverse global settings, from indigenous communities in Colombia and disaster-affected regions in Taiwan to high-school cohorts in Sweden and Finland, providing a multifaceted look at how climate awareness intersects with adolescent mental health. For each study, key statistical associations, such as odds ratios for clinical distress and longitudinal beta coefficients for behavioral engagement, are synthesized alongside qualitative conclusions to highlight the complex relationship between climate-related exposure and psychological well-being.

## Study Characteristics: Adolescent Climate Awareness and Anxiety

Summary of included studies (n=15)

| Study ID / Author | Location / Context | N | Age (Mean / Range) | Study Design | Key Statistical Findings | Qualitative Conclusions |
|---|---|---|---|---|---|---|
| **Wullenkord & Ojala (2023)** | Sweden; General high school | 795 | 16-22 | Cross-sectional | Macro worry (d=0.77); Micro worry (d=0.60); p < .001 | Climate worry rose sharply over 10 years; macro worry predicts engagement. |
| **Veijonaho et al. (2025)** | Finland; 14 schools | 684 | 12-17 | 3-wave Longitudinal | PEB to distress: β=0.44 (p<.05) | Taking pro-environmental action (PEB) appears to increase future climate distress. |
| **Park et al. (2026)** | Canada; "All Our Families" cohort | 1,230 | M=12.85 | Cross-sectional | Cognitive worry & Maternal GAD-7: r=.18 | 60% of young adolescents report worry; predicts depression independently of general anxiety. |
| **Goldberg et al. (2025)** | Bangladesh; Dhaka vs. Barisal | 1,200 | 15-18 | Mixed-methods | Depression adjOR 3.52 (High risk) | High flood exposure correlates with severe anxiety and temporal discounting. |
| **Chen et al. (2011)** | Taiwan; Typhoon Morakot displacement | 271 | M=13.4 | Cross-sectional | IES-R Cutoff 19/20; ROC Area: 0.92 | 25.8% of adolescents met PTSD diagnostic criteria 3 months post-disaster. |
| **Assaf et al. (2026)** | USA; Pediatric ED | 557 | 12-17 | Cross-sectional | Mod-sev GAD-7: ORadj 6.59 (p<.001) | Generalized anxiety is the strongest predictor of climate anxiety. |
| **Agudelo Hernández (2025)** | Colombia; Indigenous Emberá-Dobidá | 40 | M=9.23 | Mixed-methods | Territory loss & MH: r=0.761 (p<0.001) | Environmental degradation severs cultural identity; 30% risk of suicide in children. |
| **Glynn et al. (2025)** | Ireland; All-female school | 86 | 15-16 | RCT | CCS increase: 8.2; EA increase: 7.2 | Nature-based interventions boost "climate capability" and adaptive eco-anxiety levels. |
| **Léger-Goodes (2023)** | Canada; Urban/suburban families | 15 | 8-12 | Qualitative (Dyads) | Parent Mean Worry = 3.06 | Children express sadness for animals; parental awareness fosters better coping. |
| **Brown et al. (2019)** | Canada; Post-wildfire (FtMc) | 5,866 | Range 11-19 | Comparative Survey | Depression 31% vs 17% (p<.00001) | 18 months post-fire, students show significantly elevated suicidal ideation. |
| **Ge et al. (2025)** | China; Eastern region | 426 | 16-18 | 3-wave Longitudinal | T1 CCAn to T2 CCAW: β=0.208 | Awareness and anxiety form a bidirectional reinforcing feedback loop in adolescents. |
| **Thomas et al. (2022)** | USA & France | 74 | 7-18 | Qualitative | N/A | Youth report anger and betrayal by older generations; conflict over political agency. |
| **Chou et al. (2023)** | Brazil; São Paulo/Salvador | 50 | 5-18 | Qualitative | N/A | Engagement profiles tied to socioeconomic context and temporal distance of threat. |
| **Benoit et al. (2025)** | USA; Crowdsourced healthy vol. | 932 | 14-18 | RCT | Hope/Agency: p<0.001; CCA: p=0.24 (ns) | Peer videos boost hope/agency, but single exposure fails to reduce anxiety. |
| **Lerolle et al. (2025)** | France; Psychiatric patients | 87 | 12-16 | Cross-sectional | CAS to C-SSRS: β=2.58 (p=0.049) | Eco-anxiety risk disappears when controlling for clinical depression and general anxiety. |



Note: (n) reflects sample used in primary analysis.

## Quality assessment/risk of bias

The quality of the included studies was assessed using three specific tools based on study design: the Newcastle-Ottawa Scale (NOS) for observational research, the Critical Appraisal Skills Programme (CASP) checklist for qualitative studies, and the Cochrane Risk of Bias tool for randomized trials. Overall, the evidence demonstrated a high standard of quality, with most observational studies scoring "Moderate to High" and qualitative work showing strong methodological rigor. While the randomized trials presented some concerns regarding participant blinding, the reporting and selection processes remained robust across the majority of the reviewed literature. The results of the risk of bias assessment are summarized in Table 4.

++ High Quality / Low Risk + Moderate Quality - Low Quality / High Risk

Table 1. Newcastle-Ottawa Scale (NOS) Appraisal

| STUDY | SELECTION | COMPARABILITY | OUTCOME | OVERALL RATING |
|---|---|---|---|---|
| Veijanaho et al., 2025 | ++ | ++ | + | GOOD |
| Ge et al., 2025 | ++ | ++ | + | MODERATE/GOOD |
| Wullenkord & Ojala, 2023 | + | + | ++ | MODERATE/HIGH |
| Park et al., 2026 | + | + | ++ | MODERATE/HIGH |
| Lerolle et al., 2025 | + | + | ++ | MODERATE/HIGH |
| Goldberg et al., 2025 | + | + | ++ | MODERATE/HIGH |
| Chen et al., 2011 | ++ | + | ++ | HIGH |
| Assaf et al., 2026 | + | + | ++ | MODERATE/HIGH |
| Agudelo-Hernández et al., 2025 | + | - | ++ | MODERATE |
| Brown et al., 2019 | ++ | ++ | + | MODERATE/GOOD |

Table 2. CASP Appraisal (Qualitative)

| STUDY | METHODOLOGY | RECRUITMENT | ANALYSIS | OVERALL RATING |
|---|---|---|---|---|
| Thomas et al., 2022 | ++ | + | ++ | MODERATE |
| Chou et al., 2023 | ++ | + | ++ | MODERATE |
| Léger-Goodes et al., 2023 | ++ | + | ++ | GOOD |

Table 3. Cochrane Risk of Bias (RCTs)

| STUDY | SELECTION BIAS | PERFORMANCE BIAS | DETECTION BIAS | REPORTING BIAS | OVERALL RISK |
|---|---|---|---|---|---|
| Glynn et al., 2025 | ++ | - | - | ++ | + |
| Benoit et al., 2025 | + | + | ++ | ++ | + |

## Summary of included studies

This systematic review includes 15 studies investigating the psychological manifestations and functional impacts of climate-change exposure on adolescents. The studies involved youth populations with age ranges from 10 to 19 years and sample sizes ranging from small qualitative dyads of 15 participants to large-scale population surveys of 5,866 participants (out of an estimated cumulative population of 100,000)[9,12]. The primary aims included evaluating the prevalence of climate distress, exploring the bidirectional relationship between climate awareness and anxiety, and identifying how coping mechanisms and perceived institutional responses moderate mental health and behavioral outcomes.

Key outcomes measured included changes in anxiety and depression severity using validated scales like the Generalized Anxiety Disorder-7 (GAD-7), Patient Health Questionnaire-9 (PHQ-9), and the Climate Change Anxiety Scale (CCAS), alongside functional metrics such as sleep disturbance, school concentration, and temporal discounting.climate worry from generalized anxiety, identifying it as a unique predictor of pediatric depression.  These studies provide a comprehensive view of the diverse psychological responses to the climate crisis in the adolescent demographic. The findings from these studies are summarized in the following Summary Table, table 5.

Study Summary Table

Simplified overview of key findings regarding climate change and adolescent mental health.

| STUDY ID / AUTHOR | LOCATION | SAMPLE (N) | PRIMARY FINDINGS |
|---|---|---|---|
| **Veijonaho et al. (2025)** | Finland | **684** | Pro-environmental behavior (PEB) increases future distress; high distress can conversely decrease future engagement. |
| **Ge et al. (2025)** | China | **426** | A bidirectional loop exists where climate awareness increases anxiety, and higher anxiety further enhances awareness. |
| **Park et al. (2026)** | Canada | **1,230** | 60% of youth report climate worry, which uniquely predicts depression and sleep loss independently of general anxiety. |
| **Goldberg et al. (2025)** | Bangladesh | **1,200** | High flood exposure correlates with severe anxiety/depression and an increase in temporal discounting behaviors. |
| **Brown et al. (2019)** | Canada | **5,866** | 18 months post-wildfire, students showed elevated rates of depression (31%) and suicidal ideation (16%). |
| **Assaf et al. (2026)** | USA | **557** | General anxiety is the strongest predictor of climate anxiety; caregiver concerns significantly influence child anxiety levels. |
| **Glynn et al. (2025)** | Ireland | **86** | Nature-based interventions increased "climate capability" and fostered adaptive forms of eco-anxiety in teenagers. |
| **Benoit et al. (2025)** | USA | **932** | Positive peer-led videos boost hope and agency but do not significantly reduce anxiety levels in a single exposure. |
| **Lerolle et al. (2025)** | France | **87** | Eco-anxiety relates to suicide risk in isolation, but this link is mediated by underlying depression and general anxiety. |
| **Agudelo-Hernández (2025)** | Colombia | **40** | Environmental degradation severs cultural identity; 30% of children in this Indigenous community were at risk of suicide. |

*Systematic Review Summary* *Study Characteristics & Key Results • 2026*

# Discussion

This systematic review examined the relationship between climate related experiences, climate awareness, and eco-anxiety among adolescents. Across the included studies, a consistent association was observed between climate related stressors and adverse psychological outcomes, including anxiety, depressive symptoms, post traumatic stress symptoms, hopelessness, emotional distress, and suicidality [1-15]. The findings indicate that eco-anxiety in adolescents is multifactorial and may arise from both direct exposure to climate related disasters, and indirect psychological awareness of environmental degradation and future climate threat [1,2,4,8].

Importantly, eco-anxiety does not appear to be a single uniform experience. Instead, it may exist along a spectrum ranging from healthy environmental concern and increased engagement, to severe emotional distress associated with impaired functioning and poorer mental health outcomes. While some studies demonstrated that climate concern encouraged pro environmental behavior and awareness, others identified associations between higher levels of eco-anxiety, and worsening anxiety, depression, and emotional burden [1,2,4,15].

**Climate Disasters and Trauma Related Psychological Distress**

Significant psychological burden was observed following direct exposure to climate related disasters. Brown et al. (2019) reported substantially elevated rates of depression and suicidal thinking among adolescents exposed to wildfire, compared to non disaster controls [9]. Similarly, Goldberg et al. (2025) demonstrated increased odds of anxiety and depressive symptoms among adolescents living in high flood risk regions of Bangladesh [6]. Chen et al. (2011) further found that approximately one quarter of adolescents exposed to floods and mudslides met diagnostic criteria for post traumatic stress disorder three months following the disaster [10].

These findings demonstrate that environmental disasters can significantly affect adolescent mental health during a period of ongoing emotional and cognitive development. Agudelo Hernández et al. (2025) further indicated that climate related distress may also involve disruption of cultural identity, environmental belonging, and connection to ancestral land among Indigenous youth [5].

Collectively, these findings suggest that climate related disasters are associated not only with acute stress reactions, but also with prolonged emotional and psychiatric burden among adolescents.

**Climate Awareness and Anticipatory Eco Anxiety**

In addition to direct environmental exposure, several studies demonstrated that climate awareness itself may contribute to emotional distress and anxiety among young populations. Wullenkord and Ojala (2023) reported substantial increases in climate worry over a ten year period among Swedish adolescents [1], while Park et al. (2026) found that climate worry predicted depressive symptoms and sleep disruption even after accounting for generalized anxiety [8].

Longitudinal studies further supported a relationship between climate awareness and emotional distress over time. Veijanaho et al. (2025) reported that increased engagement with climate issues may contribute to greater emotional burden [4], while Ge et al. (2025) found that climate awareness and climate anxiety appeared to reinforce one another across repeated waves of adolescent follow up [2].

These findings support the concept of eco-anxiety as a chronic anticipatory stress response related to uncertainty regarding future environmental stability, and perceived lack of control. Unlike trauma related distress following disasters, anticipatory eco-anxiety appears to develop through ongoing cognitive and emotional engagement with climate change and future environmental threat.

**Adaptive Versus Maladaptive Eco Anxiety**

One notable finding was the possible dual role of eco-anxiety. Several studies suggested that moderate climate concern may function adaptively by promoting engagement, awareness, and pro environmental behavior. Wullenkord and Ojala (2023) found that “macro worry” regarding societal climate issues predicted greater climate engagement [1], while Glynn et al. (2025) demonstrated that a nature based intervention significantly increased both climate capability and adaptive eco-anxiety among adolescents [15].

However, other studies suggested that severe eco-anxiety may contribute to emotional impairment and psychiatric vulnerability. Veijanaho et al. (2025) reported that climate distress predicted lower future engagement over time, suggesting that excessive emotional burden may eventually impair adaptive functioning [4]. Lerolle et al. (2025) found that eco-anxiety was initially associated with suicide risk among hospitalized adolescent patients [7]. However, the association disappeared after controlling for generalized anxiety and depression, suggesting that eco-anxiety may occur alongside broader mental health difficulties, rather than functioning as an independent condition in adolescents with pre existing psychiatric disorders [7].

Additionally, Benoit et al. (2025) demonstrated that positive climate related peer messaging increased hope and perceived agency, but did not significantly reduce anxiety levels following single session exposure [14]. Together, these findings suggest that eco-anxiety exists along a spectrum, with moderate concern potentially encouraging positive engagement, while more severe or chronic distress may contribute to worsening mental health outcomes.

**Sociocultural and Developmental Influences**

The qualitative studies included in this review provided important insight into the lived experiences of eco-anxiety among children and adolescents. Thomas et al. (2022), Chou et al. (2023), and Léger Goodes et al. (2023) identified recurring themes of helplessness, grief, anger, uncertainty, betrayal by older generations, and concern regarding the future [11,13,12].

Importantly, these studies demonstrated that eco-anxiety is strongly influenced by social, cultural, and interpersonal context. Chou et al. (2023) found that emotional engagement with climate change varied according to socioeconomic conditions and perceived closeness to environmental threats [13]. Léger Goodes et al. (2023) and Assaf et al. (2026) highlighted the importance of parent-child and caregiver dynamics in shaping adolescents' emotional responses to climate related concerns [12,3], while Thomas et al. (2022) identified differences in perceived political agency between adolescents in the United States and France, suggesting that sociopolitical context may influence emotional responses and coping strategies related to climate change [11] .

These findings indicate that eco-anxiety in adolescents cannot be fully understood independently of developmental stage, cultural context, family systems, and environmental vulnerability.

**Methodological Strengths and Limitations**

The methodological quality of included studies ranged from moderate to high overall. Several studies utilized validated psychological scales, large sample sizes, and robust statistical analyses. The inclusion of longitudinal studies strengthened the evidence base by allowing assessment of temporal relationships between climate awareness and psychological outcomes. Additionally, qualitative studies provided important insight into emotional experiences and sociocultural influences that may not be adequately captured through quantitative measures alone.

However, several limitations should be acknowledged. Most included studies were observational and cross sectional in design, limiting the ability to determine causality. Reliance on self reported psychological outcomes was common and may increase the risk of reporting bias. Many studies also used school based, convenience, or regionally limited samples, which may limit generalizability. In addition, there were differences in how eco-anxiety, climate distress, and climate awareness were defined and measured across studies. Many studies were also conducted in Western, or high income populations, although some studies involving Indigenous communities and low and middle income countries provided important additional perspectives.

**Clinical and Public Health Implications**

Eco-anxiety represents an increasingly important mental health concern among adolescents. Climate related distress may contribute to impaired emotional functioning, chronic anxiety, depressive symptoms, maladaptive coping behaviors, and worsening psychiatric vulnerability, particularly among youth exposed to environmental disasters, or pre existing mental health challenges.

Clinicians working with adolescents should consider assessing climate related distress during mental health evaluations, particularly in disaster exposed regions and environmentally vulnerable communities. Schools and community organizations may also play an important role by promoting adaptive coping strategies, resilience building interventions, emotional processing, and positive climate engagement.

Furthermore, interventions should not aim to eliminate climate concern entirely, as moderate levels of climate awareness and emotional engagement may support adaptive coping and pro environmental behavior. Instead, interventions should focus on preventing overwhelming distress, hopelessness, and emotional impairment.

**Comparison With Existing Literature**

Previous reviews have identified associations between climate change and adverse mental health outcomes among children, adolescents, and adults, including anxiety, depression, trauma related distress, and eco-anxiety [16,17]. For example, Léger-Goodes et al. (2022) conducted a scoping review examining eco-anxiety in children and youth and highlighted important knowledge gaps regarding the psychological impacts of climate change in younger populations [17]. However, much of the existing literature focused broadly on climate change and mental health, or on describing the emerging concept of eco-anxiety, rather than examining eco-anxiety among adolescents using more recent longitudinal, qualitative, and interventional evidence. In contrast, the present review focused specifically on children and adolescents, and included quantitative, qualitative, longitudinal, and interventional studies. This review also examined both direct climate related exposures, and anxiety related to awareness of future environmental threats [1,2,4,8]. Additionally, the findings of this review suggest that eco-anxiety may function both as an adaptive response that encourages environmental engagement, and as a source of significant emotional distress [1,2,4,15]. As a result, this review provides a broader and more comprehensive understanding of eco-anxiety in adolescents, and addresses important gaps in the existing literature.

**Future Research Directions**

Future research should prioritize longitudinal and interventional study designs to better understand the relationship between climate awareness, environmental exposure, and psychological outcomes over time. More standardized definitions and measurement tools for eco-anxiety are also needed to improve consistency and comparability across studies.

Further research involving low and middle income countries, Indigenous communities, and adolescents with pre existing psychiatric conditions remains important, as these groups may experience different vulnerabilities and coping responses. Studies using clinician administered assessments, and more objective mental health measures may also help reduce reliance on self reported outcomes.

# Conclusion

Overall, eco-anxiety is a significant and increasingly important mental health concern among adolescents. Both direct exposure to climate related disasters, and indirect awareness of environmental threat may contribute to emotional distress and adverse psychological outcomes. However, eco-anxiety may function along a spectrum, ranging from adaptive climate concern and engagement in pro environmental behaviors, to clinically significant emotional impairment. Although the included studies varied in design and methodology, the overall consistency of findings indicates that eco-anxiety is an important and growing public health concern that deserves continued clinical, psychological, and policy attention.

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